

# Competition – Contractor Experience

2024-08-22

- Let  $n_i^{task}$  = Number of projects of the same task held by the contractor in the three quarters before starting project  $i$
- Let  $n_i$  = Number of projects held by the contractor in the three quarters before starting project  $i$
- Define:
  - Inexperienced contractor $_i$  = 1 if  $n_i^{task} = 0$
  - New entrant $_i$  = 1 if  $n_i = 0$
  - Level of inexperience $_i$  =  $1 - \frac{n_i^{task}}{n_i}$  if  $n_i > 0$
  - Level of inexperience $_i$  = 1 if  $n_{i,k,t} = 0$ .

## 1 Regression results

- For  $IE + Treat \times IE + IE \times Post$  using the two discrete metrics, we get:

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|                    | Estimate | SE   | 2.5 % | 97.5 % | Hypothesis | z value | Pr(> z ) |
|--------------------|----------|------|-------|--------|------------|---------|----------|
| Task inexperienced | 1.04     | 0.15 | 0.74  | 1.34   | 0.00       | 6.83    | 0.00     |

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|             | Estimate | SE   | 2.5 % | 97.5 % | Hypothesis | z value | Pr(> z ) |
|-------------|----------|------|-------|--------|------------|---------|----------|
| New entrant | 1.02     | 0.21 | 0.61  | 1.43   | 0.00       | 4.85    | 0.00     |

Table 1: Logit model

|                       | IE                  |                     |                       |
|-----------------------|---------------------|---------------------|-----------------------|
|                       | Task inexperienced  | New entrant         | Level of inexperience |
|                       | I                   | II                  | III                   |
|                       | Logit               | Logit               | OLS                   |
| Treat                 | 0.407***<br>(0.032) | 0.556***<br>(0.042) | 0.001<br>(0.003)      |
| Treat $\times$ Post   | 0.083***<br>(0.032) | 0.229***<br>(0.041) | 0.016***<br>(0.004)   |
| Time FE               | Yes                 | Yes                 | Yes                   |
| Task FE               | Yes                 | Yes                 | Yes                   |
| Industry FE           | Yes                 | Yes                 | Yes                   |
| Observations          | 182,794             | 189,340             | 192,743               |
| Squared Correlation   | 0.30                | 0.20                | 0.31                  |
| Pseudo R <sup>2</sup> | 0.25                | 0.22                | 0.45                  |
| BIC                   | 177,101.39          | 125,903.49          | 104,731.94            |

*Clustered (Project ID) standard-errors in parentheses*

*Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1*

|                                 | Percentage delay rate |                      |                       |
|---------------------------------|-----------------------|----------------------|-----------------------|
|                                 | Task inexperienced    | New entrant          | Level of inexperience |
|                                 | I                     | II                   | III                   |
| Treat                           | -1.007***<br>(0.139)  | -1.006***<br>(0.118) | -0.188<br>(0.263)     |
| IE                              | 1.084***<br>(0.178)   | 0.873***<br>(0.240)  | 2.172***<br>(0.253)   |
| Treat $\times$ IE               | -0.151<br>(0.204)     | -0.255<br>(0.293)    | -1.063***<br>(0.309)  |
| IE $\times$ Post                | -0.432*<br>(0.256)    | -0.118<br>(0.362)    | -0.771**<br>(0.321)   |
| Treat $\times$ Post             | 1.014***<br>(0.173)   | 1.085***<br>(0.152)  | 0.667**<br>(0.319)    |
| Treat $\times$ IE $\times$ Post | 0.540*<br>(0.310)     | 0.521<br>(0.449)     | 0.553<br>(0.403)      |
| Time FE                         | Yes                   | Yes                  | Yes                   |
| Task FE                         | Yes                   | Yes                  | Yes                   |
| Industry FE                     | Yes                   | Yes                  | Yes                   |
| Observations                    | 192,629               | 192,629              | 192,629               |
| R <sup>2</sup>                  | 0.27                  | 0.27                 | 0.27                  |
| Within R <sup>2</sup>           | 0.19                  | 0.19                 | 0.19                  |

*Clustered (Project ID) standard-errors in parentheses*  
*Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1*